

YEAR 7 MATHEMATICS

Chapter 12

Ratios, rates and time

MIXED PRACTICE WORKSHEETS

12

How to use this booklet

- Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

WE DO

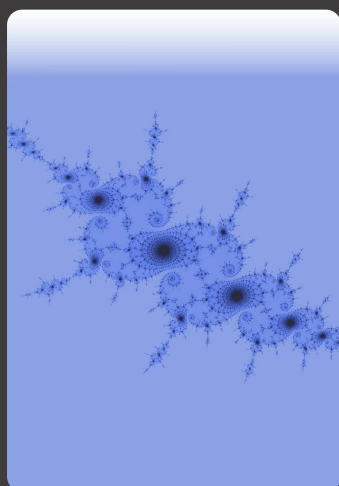
The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
12.01	Ratios	3		☐
12.02	Equivalent and simplified ratios	5		☐
12.03	Ratio problems	7		☐
12.04	Rates	9		☐
12.05	Best buys	11		☐
12.06	Rate problems	13		☐
12.07	Travel graphs	15		☐
12.08	Time	17		☐
12.09	12-hour and 24-hour time	19		☐
12.10	Time differences	21		☐
12.11	Timetables	23		☐

In this chapter you will

- write, simplify and use ratios
- solve ratio problems
- write rates and find the best buy
- solve rate problems and read travel graphs
- round, convert and calculate with time
- use 12-hour and 24-hour time, time differences and timetables



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a jellyfish of spirals near the neck of the left bulb. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME _____ CLASS _____ DATE _____

WE DO
KEY IDEA

A ratio compares quantities in a set order: 3 : 5 means 3 parts of the first to 5 parts of the second. The order matters. With 3 : 5, the first is $\frac{3}{8}$ of the whole.

Watch your teacher.
Copy every step.

- E1** A bag has 4 red and 7 blue counters. Write the ratio of red to blue.

- E2** Write the ratio of shaded to unshaded squares.



- E3** In a ratio of 3 : 5, what fraction is the first part?

Easy

Questions 1–8

- 1** A class has 9 boys and 12 girls. Write the ratio of girls to boys.

- 2** Write the ratio of shaded to unshaded squares.



- 3** In a ratio of 6 : 5, what fraction is the first part?

- 4** A class has 10 boys and 11 girls. Write the ratio of girls to boys.

- 5** Write the ratio of shaded to unshaded squares.



- 6** In a ratio of 8 : 4, what fraction is the first part?

- 7** A class has 5 boys and 2 girls. Write the ratio of girls to boys.

- 8** Write the ratio of shaded to unshaded squares.


Easy

Questions 9–16

- 9** Write the ratio of shaded to unshaded squares.



- 10** In a ratio of 9 : 8, what fraction is the first part?

- 11** A class has 2 boys and 4 girls. Write the ratio of girls to boys.

- 12** Find the area of the triangular end: base 4 cm, height 3 cm.

from 8.12

- 13** A fruit bowl has 7 apples, 9 pears and 7 plums. Write the ratio of plums to apples to pears.

- 14** Write the ratio of shaded to unshaded squares.



- 15** In a ratio of 3 : 1, what fraction is the first part?

- 16** Write $\frac{14}{8}$ as a mixed numeral.

from 4.01

Medium

Questions 17–23

- 17** A fruit bowl has 3 apples, 2 pears and 4 plums. Write the ratio of plums to apples to pears.

- 18** A team won 11 games and lost 15. Write the ratio of wins to games played.

- 19** Write $\frac{25}{3}$ as a mixed numeral.

from 4.01

- 20** A fruit bowl has 8 apples, 6 pears and 4 plums. Write the ratio of plums to apples to pears.

- 21** A team won 12 games and lost 10. Write the ratio of wins to games played.

- 22** Set A: 11, 18, 12, 2, 20. Set B: 3, 14, 3, 8, 15. Which set has the higher median?

from 10.08

- 23** A fruit bowl has 4 apples, 5 pears and 5 plums. Write the ratio of plums to apples to pears.

Medium

Questions 24–32

- 24 A team won 7 games and lost 8. Write the ratio of wins to games played.

- 25 A fruit bowl has 3 apples, 7 pears and 3 plums. Write the ratio of plums to apples to pears.

- 26 Write $3\frac{4}{5}$ as an improper fraction.

from 4.01

- 27 Red and blue paint are mixed in the ratio 2 : 5. What fraction of the mix is blue? How many litres of blue are in 42 L?

- 28 Simplify $\frac{12}{72}$ fully.

from 4.01

- 29 A team won 12 games and lost 13. Write the ratio of wins to games played.

- 30 A fruit bowl has 3 apples, 6 pears and 8 plums. Write the ratio of plums to apples to pears.

- 31 Does (4, 11) lie on the line $y = x + 5$? Show the check.

from 9.06

- 32 Red and blue paint are mixed in the ratio 5 : 3. What fraction of the mix is blue? How many litres of blue are in 48 L?

Challenging

Questions 33–40

- 33 Red and blue paint are mixed in the ratio 4 : 2. What fraction of the mix is blue? How many litres of blue are in 30 L?

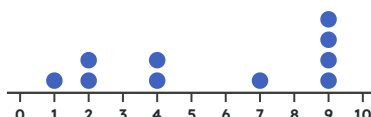
- 34 Tom has 5 pens and Zac has 9. Is the ratio 5 : 9 the same as 9 : 5? Explain.

- 35 Write $\frac{79}{8}$ as a mixed numeral.

from 4.01

- 36 Red and blue paint are mixed in the ratio 4 : 3. What fraction of the mix is blue? How many litres of blue are in 35 L?

- 37 Find the mode.



from 10.07

- 38 Noah has 5 pens and Jack has 4. Is the ratio 5 : 4 the same as 4 : 5? Explain.

- 39 Red and blue paint are mixed in the ratio 5 : 3. What fraction of the mix is blue? How many litres of blue are in 40 L?

- 40 Write $\frac{29}{8}$ as a mixed numeral.

from 4.01

Challenging

Questions 41–47

- 41 Zac has 9 pens and Aisha has 8. Is the ratio 9 : 8 the same as 8 : 9? Explain.

- 42 Red and blue paint are mixed in the ratio 2 : 4. What fraction of the mix is blue? How many litres of blue are in 24 L?

- 43 How many values are 47 or more?

Stem	Leaf
3	1 9
4	8 9
5	0 5
6	0 3
7	1 5

Key: 3 | 5 = 35

from 10.04

- 44 A team won 15 games and lost 8. Write the ratio of wins to games played.

- 45 Write $6\frac{5}{8}$ as an improper fraction.

from 4.01

- 46 Kai has 7 pens and Mia has 8. Is the ratio 7 : 8 the same as 8 : 7? Explain.

- 47 Red and blue paint are mixed in the ratio 3 : 5. What fraction of the mix is blue? How many litres of blue are in 16 L?

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Equivalent ratios: multiply or divide both parts by the same number. To simplify, divide both parts by their HCF. Change to the same units first, and remove decimals by multiplying.

E1 Simplify 12 : 18.

E2 Simplify 40 cm : 2 m.

E3 Find the missing number:
 $3 : 4 = \square : 20$.

Easy

Questions 1–9

1 Simplify 5 : 10.

2 Find the missing number: 5 :
 $4 = \square : 20$.

3 Write two ratios equivalent to
1 : 9.

4 Simplify 56 : 24.

5 Find the missing number: 9 :
 $8 = \square : 56$.

6 Write two ratios equivalent to
1 : 5.

7 Simplify 40 : 5.

8 Find the missing number: 3 :
 $8 = \square : 64$.

9 Write the ratio of shaded to
unshaded squares.



from 12.01

Easy

Questions 10–18

10 Find the missing number: 8 :
 $5 = \square : 30$.

11 Write two ratios equivalent to
7 : 8.

12 Simplify 36 : 42.

13 In the table, what is y when
 $x = 2$?

x	1	2	3	4
y	5	8	11	14

from 9.04

14 Simplify 24 : 4.

HCF is ____

= ____ : ____

15 Find the missing number: 7 :
 $1 = \square : 6$.

16 Write two ratios equivalent to
3 : 8.

17 Is 4 a common factor of 24
and 17?

from 3.11

18 Simplify 7 : 49.

Medium

Questions 19–26

19 Simplify 40 : 24.

20 Simplify 200 mL : 2 L.

21 A fruit bowl has 8 apples, 5
pears and 7 plums. Write the
ratio of plums to apples
to pears.

from 12.01

22 Simplify 1 : 0.5.

23 Simplify 105 : 90.

24 A letter is chosen from the
word SCHOOL. List the
sample space.

from 11.01

25 Simplify 10 cm : 2 m.

26 Simplify 1.5 : 2.

Medium

Questions 27–37

- 27 Simplify 250 g : 3 kg.

- 28 Simplify 4.5 : 4.

- 29 List the common factors of 20 and 12.

from 3.11

- 30 Simplify 36 : 96.

- 31 A team won 4 games and lost 13. Write the ratio of wins to games played.

from 12.01

- 32 Simplify 10 : 25 : 15.

- 33 Simplify 25 cm : 2 m.

- 34 Start at C3. Move 2 left and 1 down. What is the grid reference now?

from 9.01

- 35 Simplify 3 : 2.5.

- 36 Find the HCF of 12 and 6.

from 3.11

- 37 Simplify 72 : 40.

Challenging

Questions 38–47

- 38 Simplify 16 : 8 : 12.

- 39 Simplify
- $\frac{1}{5} : \frac{1}{2}$
- .

- 40 A fruit bowl has 8 apples, 6 pears and 8 plums. Write the ratio of plums to apples to pears.

from 12.01

- 41 Priya says
- $5 : 9 = 9 : 13$
- because you add the same to both. Is that right? Explain.

- 42 Where does
- $y = x + 2$
- cross the y-axis?

from 9.06

- 43 Simplify 3 : 6 : 9.

- 44 Simplify
- $\frac{3}{5} : \frac{1}{2}$
- .

- 45 Is 5 a common factor of 30 and 19?

from 3.11

- 46 Kai says
- $4 : 9 = 7 : 12$
- because you add the same to both. Is that right? Explain.

- 47 A team won 14 games and lost 5. Write the ratio of wins to games played.

from 12.01

Challenging

Questions 48–56

- 48 Simplify
- $\frac{1}{3} : \frac{1}{2}$
- .

- 49 Simplify 4 : 6 : 10.

- 50 A card is picked from a full deck of 52. Find P(a picture card (J, Q, K)). Write it as a fraction and a percentage.

from 11.02

- 51 Simplify 2.5 : 3.

- 52 List the common factors of 8 and 20.

from 3.11

- 53 Simplify
- $\frac{1}{4} : \frac{1}{2}$
- .

- 54 Simplify 15 : 25 : 10.

- 55 A fruit bowl has 8 apples, 9 pears and 9 plums. Write the ratio of plums to apples to pears.

from 12.01

- 56 Simplify 1.5 : 4.5.

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

To share in a ratio, add the parts, find one part (total ÷ parts), then multiply. If you know one quantity, use it to find what one part is worth.

E1 Share \$60 in the ratio 2 : 3.

E2 Boys to girls is 4 : 5. There are 20 boys. How many girls?

E3 Cordial and water are mixed 1 : 6. How much cordial for 2.1 L of drink?

Easy

Questions 1–9

1 Share \$90 in the ratio 5 : 4.

2 Boys to girls is 5 : 3. There are 30 boys. How many girls?

3 Share \$18 in the ratio 2 : 1.

4 Boys to girls is 4 : 1. There are 20 boys. How many girls?

5 Share \$12 in the ratio 2 : 1.

6 Boys to girls is 2 : 5. There are 8 boys. How many girls?

7 Share \$36 in the ratio 3 : 1.

8 Boys to girls is 3 : 4. There are 15 boys. How many girls?

9 Find the missing number: 3 : 4 = □ : 12.

from 12.02

Easy

Questions 10–17

10 Boys to girls is 1 : 3. There are 2 boys. How many girls?

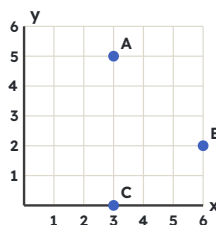
11 Share \$20 in the ratio 2 : 3.

12 Share 56 lollies in the ratio 2 : 2.

___ parts: one part = ___

___ and ___

13 Which point is at (3, 0)?



from 9.02

14 Boys to girls is 3 : 4. There are 6 boys. How many girls?

15 Share \$24 in the ratio 4 : 4.

16 Share 126 lollies in the ratio 1 : 5.

17 Write the ratio of shaded to unshaded squares.



from 12.01

Medium

Questions 18–24

18 Share 104 lollies in the ratio 5 : 3.

19 Cordial and water are mixed 2 : 4. How much water goes with 500 mL of cordial?

20 Simplify 1 : 3.

from 12.02

21 Chloe and Jack share money in the ratio 2 : 4. Chloe gets \$10. How much does Jack get?

22 Share 56 lollies in the ratio 1 : 3.

23 A bag has 1 blue and 2 white counters. One is picked. Find P(blue).

from 11.02

24 Cordial and water are mixed 4 : 7. How much water goes with 600 mL of cordial?

Medium

Questions 25–32

- 25 Cordial and water are mixed 3 : 9. How much water goes with 900 mL of cordial?

- 26 Tom and Kai share money in the ratio 5 : 4. Tom gets \$15. How much does Kai get?

- 27 In a ratio of 6 : 5, what fraction is the first part?

from 12.01

- 28 Share 168 lollies in the ratio 7 : 1.

- 29 Simplify 18 : 81.

from 12.02

- 30 Share \$18 in the ratio 1 : 2 : 3.

- 31 Cordial and water are mixed 5 : 6. How much water goes with 500 mL of cordial?

- 32 How many values are 50 or more?

Stem	Leaf
3	9
4	6 6
5	6 6 9
6	2 3
7	1 7

Key: 3 | 5 = 35

from 10.04

Challenging

Questions 33–41

- 33 Share \$56 in the ratio 1 : 3 : 4.

- 34 The ratio of boys to girls is 5 : 4. There are 5 more boys than girls. How many students are there?

- 35 Simplify 50 cm : 2 m.

from 12.02

- 36 A recipe uses flour and sugar in the ratio 7 : 4. You have 2000 g of sugar. How much flour do you need?

- 37 In the table, what is y when $x = 2$?

x	1	2	3	4
y	9	13	17	21

from 9.04

- 38 Share \$28 in the ratio 2 : 4 : 1.

- 39 The ratio of boys to girls is 2 : 5. There are 15 more girls than boys. How many students are there?

- 40 Write the ratio of shaded to unshaded squares.



from 12.01

- 41 A recipe uses flour and sugar in the ratio 7 : 2. You have 1000 g of sugar. How much flour do you need?

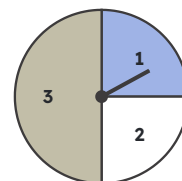
Challenging

Questions 42–49

- 42 The ratio of boys to girls is 2 : 5. There are 12 more girls than boys. How many students are there?

- 43 A recipe uses flour and sugar in the ratio 6 : 3. You have 1200 g of sugar. How much flour do you need?

- 44 Are the outcomes on this spinner equally likely?



from 11.01

- 45 Share \$56 in the ratio 4 : 3 : 1.

- 46 In a ratio of 4 : 3, what fraction is the first part?

from 12.01

- 47 Lena and Jack share money in the ratio 5 : 4. Lena gets \$20. How much does Jack get?

- 48 The ratio of boys to girls is 2 : 3. There are 4 more girls than boys. How many students are there?

- 49 Simplify 30 : 15.

from 12.02

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

A rate compares two different kinds of quantity, such as km and hours. Write the units, then divide to find the rate for one unit: 240 km in 3 h is 80 km/h.

E1 A car travels 240 km in 3 hours. Find its speed.

E2 4 kg of apples cost \$12. Find the cost per kilogram.

E3 A tap fills 90 L in 6 minutes. Write the rate in L/min.

Easy

Questions 1–9

1 A car travels 297 km in 3 hours. Find its speed.

2 5 kg of apples cost \$30. Find the cost per kilogram.

3 A tap fills 54 L in 6 minutes. Write the rate in L/min.

4 A car travels 360 km in 6 hours. Find its speed.

5 7 kg of apples cost \$63. Find the cost per kilogram.

6 A tap fills 42 L in 2 minutes. Write the rate in L/min.

7 A car travels 735 km in 7 hours. Find its speed.

8 9 kg of apples cost \$81. Find the cost per kilogram.

9 Find the missing number: 5 : 8 = $\square$: 16.

from 12.02

Easy

Questions 10–18

10 9 kg of apples cost \$27. Find the cost per kilogram.

11 A tap fills 28 L in 2 minutes. Write the rate in L/min.

12 A car travels 483 km in 7 hours. Find its speed.

13 Find the area of the triangular end: base 6 cm, height 7 cm.

from 8.12

14 Mia earns \$78 for 6 hours of work. Find the hourly rate.

$$\underline{\hspace{1cm}} \div \underline{\hspace{1cm}} \text{ h}$$

$$= \underline{\hspace{1cm}} / \text{h}$$

15 8 kg of apples cost \$64. Find the cost per kilogram.

16 A tap fills 126 L in 6 minutes. Write the rate in L/min.

17 $26.67 \div 3$

from 7.07

18 A car travels 138 km in 2 hours. Find its speed.

Medium

Questions 19–24

19 Noah earns \$65 for 5 hours of work. Find the hourly rate.

20 A heart beats 252 times in 4 minutes. Find the rate in beats per minute.

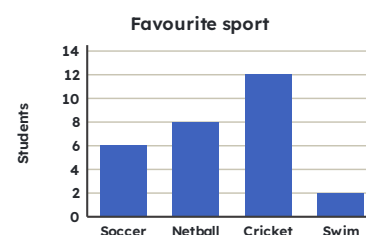
21 Cordial and water are mixed 2 : 8. How much water goes with 600 mL of cordial?

from 12.03

22 A car uses 31.5 L of petrol for 350 km. Write the rate in L per 100 km.

23 Chloe earns \$95 for 5 hours of work. Find the hourly rate.

24 Which sport was the most popular?



from 10.01

Medium

Questions 25–34

- 25 A heart beats 162 times in 2 minutes. Find the rate in beats per minute.

- 26 A car uses 28.8 L of petrol for 360 km. Write the rate in L per 100 km.

- 27 Write the ratio of shaded to unshaded squares.



from 12.01

- 28 Tom earns \$60 for 4 hours of work. Find the hourly rate.

- 29 Simplify 3.5 : 2.5.

from 12.02

- 30 Change 50 m/s to km/h.

- 31 A heart beats 156 times in 2 minutes. Find the rate in beats per minute.

- 32 Change 8 m^3 to cm^3 .

from 8.10

- 33 A car uses 12.9 L of petrol for 430 km. Write the rate in L per 100 km.

- 34 $4.5 \div 5$

from 7.07

Challenging

Questions 35–44

- 35 Change 60 m/s to km/h.

- 36 Change 108 km/h to m/s.

- 37 Jack and Noah share money in the ratio 2 : 3. Jack gets \$10. How much does Noah get?

from 12.03

- 38 500 g of cheese costs \$8. Find the cost per kilogram.

- 39 A letter is chosen from the word SUMMER. List the sample space.

from 11.01

- 40 Change 30 m/s to km/h.

- 41 Change 90 km/h to m/s.

- 42 In a ratio of 6 : 4, what fraction is the first part?

from 12.01

- 43 200 g of cheese costs \$9. Find the cost per kilogram.

- 44 Simplify 60 : 75.

from 12.02

Challenging

Questions 45–54

- 45 Change 72 km/h to m/s.

- 46 200 g of cheese costs \$7. Find the cost per kilogram.

- 47 Which rule fits the table:
 $y = 4x + 4$, or $y = 3x + 5$, or $y = 5x + 3$?

x	1	2	3
y	8	11	14

from 9.05

- 48 Change 40 m/s to km/h.

- 49 $13.3 \div 7$

from 7.07

- 50 A car uses 18.5 L of petrol for 370 km. Write the rate in L per 100 km.

- 51 250 g of cheese costs \$7. Find the cost per kilogram.

- 52 Share 84 lollies in the ratio 2 : 2.

from 12.03

- 53 Change 20 m/s to km/h.

- 54 Set A: 16, 17, 16, 15, 14. Set B: 11, 17, 5, 13, 2. Which set has the higher median?

from 10.08

NAME _____ CLASS _____ DATE _____

WE DO
KEY IDEA

To find the best buy, find the unit price of each (the cost for 1 g, 100 g or 1 item), then compare. The lowest unit price is the best value.

Watch your teacher.
Copy every step.

E1 500 g for \$4 or 750 g for \$5.25. Which is the better buy?

E2 Find the cost of 1 pen when 6 pens cost \$4.50.

E3 A 2 L milk is \$3.60 and a 3 L milk is \$5.10. Which is better value?

Easy

Questions 1–9

1 8 pens cost \$3.20. Find the cost of 1 pen.

2 4 kg costs \$12.00 or 6 kg costs \$15.00. Find the price per kg of each.

3 6 pens cost \$1.62. Find the cost of 1 pen.

4 4 kg costs \$16.00 or 6 kg costs \$21.00. Find the price per kg of each.

5 8 pens cost \$7.04. Find the cost of 1 pen.

6 3 kg costs \$15.00 or 6 kg costs \$33.00. Find the price per kg of each.

7 4 pens cost \$0.96. Find the cost of 1 pen.

8 4 kg costs \$24.00 or 5 kg costs \$32.50. Find the price per kg of each.

9 9 kg of apples cost \$63. Find the cost per kilogram.

from 12.04

Easy

Questions 10–15

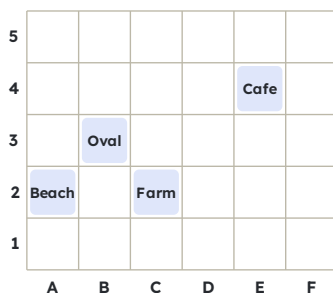
10 3 kg costs \$15.00 or 4 kg costs \$22.00. Find the price per kg of each.

11 4 pens cost \$1.80. Find the cost of 1 pen.

12 250 g for \$1.98 or 1000 g for \$7.20. Which is the better buy?

___ g is better

13 Give the grid reference of the Beach.



from 9.01

14 5 kg costs \$25.00 or 6 kg costs \$33.00. Find the price per kg of each.

15 4 pens cost \$3.32. Find the cost of 1 pen.

Medium

Questions 16–21

16 400 g for \$3.08 or 600 g for \$4.98. Which is the better buy?

17 A 2 L bottle costs \$4.90 and a 4 L bottle costs \$8.88. Which is better value?

18 A team won 4 games and lost 15. Write the ratio of wins to games played.

from 12.01

19 250 g for \$2.30 or 1000 g for \$8.20. Which is the better buy?

20 A 2 L bottle costs \$3.70 and a 3 L bottle costs \$4.80. Which is better value?

21 Cordial and water are mixed 4 : 9. How much water goes with 1000 mL of cordial?

from 12.03

Medium

Questions 22–31

- 22 A 1 L bottle costs \$1.59 and a 3 L bottle costs \$4.62. Which is better value?

- 23 300 g for \$2.64 or 500 g for \$4.85. Which is the better buy?

- 24 A die is rolled 60 times. How many 6s would you expect?

from 11.04

- 25 Muffins are \$5 each, or 4 for \$19. How much do you save on 8 muffins with the deal?

- 26 In a ratio of 9 : 3, what fraction is the first part?

from 12.01

- 27 A 1 L bottle costs \$1.76 and a 2 L bottle costs \$3.06. Which is better value?

- 28 400 g for \$2.76 or 600 g for \$3.72. Which is the better buy?

- 29 Find the missing number: $2 : 1 = \square : 7$.

from 12.02

- 30 Muffins are \$3 each, or 4 for \$11. How much do you save on 8 muffins with the deal?

- 31 A car uses 22.5 L of petrol for 450 km. Write the rate in L per 100 km.

from 12.04

Challenging

Questions 32–40

- 32 Muffins are \$3 each, or 3 for \$8. How much do you save on 6 muffins with the deal?

- 33 A 3-pack costs \$5.40 and a 6-pack costs \$11.00. Which is better value? Explain.

- 34 A box has a volume of 162 cm^3 . It is 3 cm long and 6 cm wide. Find its height.

from 8.11

- 35 Muffins are \$2 each, or 3 for \$5. How much do you save on 6 muffins with the deal?

- 36 $4.4 \div 0.1$

from 7.08

- 37 A 5-pack costs \$3.00 and a 10-pack costs \$5.90. Which is better value? Explain.

- 38 Muffins are \$4 each, or 4 for \$15. How much do you save on 8 muffins with the deal?

- 39 A team won 12 games and lost 13. Write the ratio of wins to games played.

from 12.01

- 40 A 3-pack costs \$4.10 and a 6-pack costs \$8.40. Which is better value? Explain.

Challenging

Questions 41–49

- 41 A 3-pack costs \$3.80 and a 6-pack costs \$7.80. Which is better value? Explain.

- 42 Muffins are \$4 each, or 3 for \$11. How much do you save on 6 muffins with the deal?

- 43 A graph has no labels on its axes. Why is that a problem?

from 10.02

- 44 A 1 L bottle costs \$2.49 and a 3 L bottle costs \$7.23. Which is better value?

- 45 Write the ratio of shaded to unshaded squares.



from 12.01

- 46 A 6-pack costs \$5.40 and a 12-pack costs \$10.60. Which is better value? Explain.

- 47 Muffins are \$2 each, or 5 for \$9. How much do you save on 10 muffins with the deal?

- 48 Write two ratios equivalent to $4 : 5$.

from 12.02

- 49 A 2 L bottle costs \$3.42 and a 3 L bottle costs \$4.35. Which is better value?

NAME _____ CLASS _____ DATE _____

WE DO
KEY IDEA

Use the rate to work out the rest: distance = speed $\times$ time, and time = distance $\div$ speed. Cost = rate $\times$ amount. Keep the units the same.

Watch your teacher.
Copy every step.

E1 A car travels at 80 km/h for 3 hours. How far does it go?

E2 How long does 150 km take at 60 km/h?

E3 Petrol is \$1.90/L. Find the cost of 40 L.

Easy

Questions 1–9

1 A car travels at 47 km/h for 2 hours. How far does it go?

2 Petrol is \$1.62 a litre. Find the cost of 36 L.

3 Kai earns \$16 an hour. How much is earned in 5 hours?

4 A car travels at 69 km/h for 4 hours. How far does it go?

5 Petrol is \$1.75 a litre. Find the cost of 55 L.

6 Jack earns \$12 an hour. How much is earned in 6 hours?

7 A car travels at 89 km/h for 4 hours. How far does it go?

8 Petrol is \$2.14 a litre. Find the cost of 28 L.

9 5 kg of apples cost \$40. Find the cost per kilogram.

from 12.04

Easy

Questions 10–18

10 Petrol is \$2.12 a litre. Find the cost of 45 L.

11 Chloe earns \$15 an hour. How much is earned in 5 hours?

12 A car travels at 65 km/h for 3 hours. How far does it go?

13 How many faces does a rectangular prism have?

from 8.09

14 How long does a 200 km trip take at 50 km/h?

____ $\div$ ____ = ____

= ____ h

15 Petrol is \$1.93 a litre. Find the cost of 59 L.

16 Priya earns \$15 an hour. How much is earned in 6 hours?

17 5 kg costs \$10.00 or 6 kg costs \$9.00. Find the price per kg of each.

from 12.05

18 A car travels at 85 km/h for 5 hours. How far does it go?

Medium

Questions 19–25

19 How long does a 100 km trip take at 40 km/h?

20 A tap runs at 16 L/min. How long does it take to fill a 64 L tank?

21 A team won 10 games and lost 8. Write the ratio of wins to games played.

from 12.01

22 A cyclist rides at 20 km/h for 15 minutes. How far is that?

23 How long does a 20 km trip take at 40 km/h?

24 Cordial and water are mixed 2 : 4. How much water goes with 600 mL of cordial?

from 12.03

25 A tap runs at 23 L/min. How long does it take to fill a 138 L tank?

Medium

Questions 26–35

- 26 A tap runs at 14 L/min. How long does it take to fill a 56 L tank?

- 27 A cyclist rides at 24 km/h for 20 minutes. How far is that?

- 28 Where does $y = x + 3$ cross the y -axis?

from 9.06

- 29 How long does a 300 km trip take at 100 km/h?

- 30 Find the missing number: $3 : 2 = \square : 16$.

from 12.02

- 31 A car goes 66 km/h for 3 h, then 82 km/h for 1 h. Find the total distance and the average speed.

- 32 A tap runs at 15 L/min. How long does it take to fill a 105 L tank?

- 33 A tap fills 20 L in 2 minutes. Write the rate in L/min.

from 12.04

- 34 A cyclist rides at 20 km/h for 20 minutes. How far is that?

- 35 400 g for \$3.88 or 600 g for \$5.28. Which is the better buy?

from 12.05

Challenging

Questions 36–43

- 36 A car goes 56 km/h for 3 h, then 93 km/h for 1 h. Find the total distance and the average speed.

- 37 A car uses 8 L per 100 km. Petrol costs \$1.99/L. Find the petrol cost for a 350 km trip.

- 38 A card is picked from a full deck of 52. Find $P(\text{a king})$. Write it as a fraction and a percentage.

from 11.02

- 39 A car goes 67 km/h for 3 h, then 88 km/h for 1 h. Find the total distance and the average speed.

- 40 Share \$25 in the ratio 1 : 4.

from 12.03

- 41 A car uses 7 L per 100 km. Petrol costs \$2.00/L. Find the petrol cost for a 440 km trip.

- 42 A car goes 57 km/h for 2 h, then 84 km/h for 2 h. Find the total distance and the average speed.

- 43 A fruit bowl has 7 apples, 7 pears and 5 plums. Write the ratio of plums to apples to pears.

from 12.01

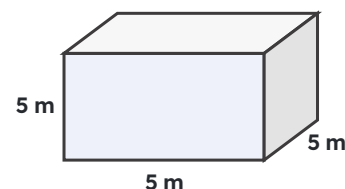
Challenging

Questions 44–51

- 44 A car uses 7 L per 100 km. Petrol costs \$1.95/L. Find the petrol cost for a 550 km trip.

- 45 A car goes 66 km/h for 3 h, then 94 km/h for 1 h. Find the total distance and the average speed.

- 46 Find the volume.



from 8.11

- 47 A cyclist rides at 16 km/h for 30 minutes. How far is that?

- 48 Write two ratios equivalent to $5 : 3$.

from 12.02

- 49 A car uses 6 L per 100 km. Petrol costs \$1.83/L. Find the petrol cost for a 550 km trip.

- 50 A car goes 61 km/h for 2 h, then 99 km/h for 1 h. Find the total distance and the average speed.

- 51 4 kg costs \$16.00 or 5 kg costs \$22.50. Find the price per kg of each.

from 12.05

NAME _____ CLASS _____ DATE _____

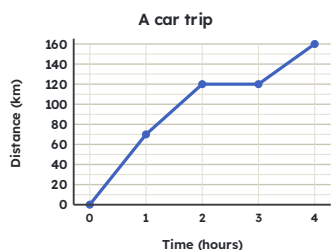
WE DO

Watch your teacher.
Copy every step.

KEY IDEA

A travel graph shows distance against time. A steeper line is faster; a flat line means stopped. Speed = distance travelled ÷ time taken for that part.

E1 How far from home is the car after 2 hours?



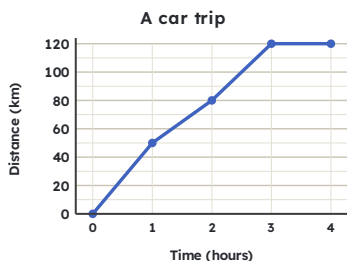
E2 When did the car stop? For how long?

E3 Find the speed in the first hour.

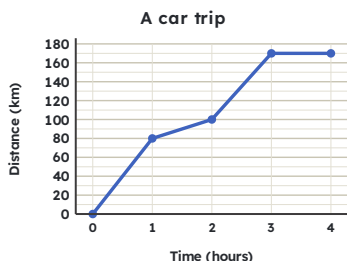
Easy

Questions 1–3

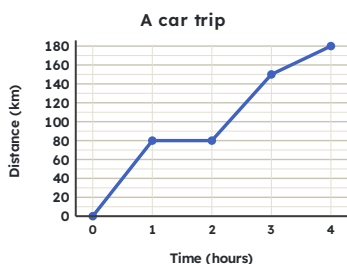
1 How far from home is the car after 1 hour?



2 During which hour did the car stop?



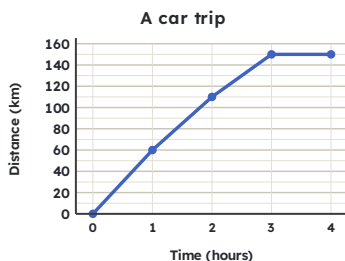
3 How far from home is the car after 1 hour?



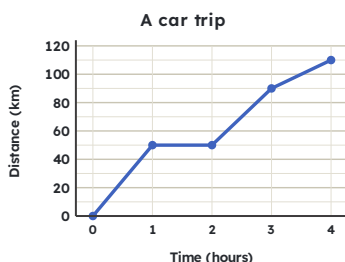
Easy

Questions 4–6

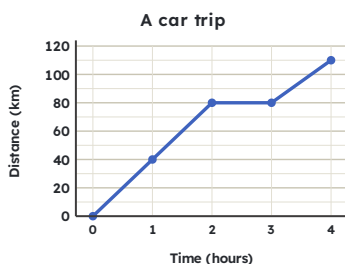
4 During which hour did the car stop?



5 How far from home is the car after 3 hours?



6 Find the speed between 1 h and 2 h.



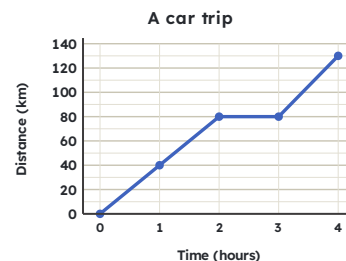
$80 - 40 = \underline{\hspace{2cm}}$

$\underline{\hspace{1cm}} \text{ km/h}$

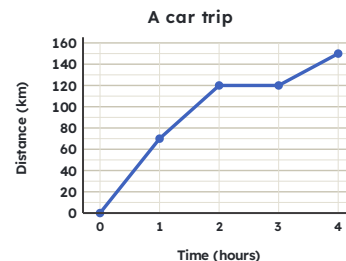
Medium

Questions 7–9

7 Find the speed between 3 h and 4 h.



8 In which hour was the car fastest? How can you tell?



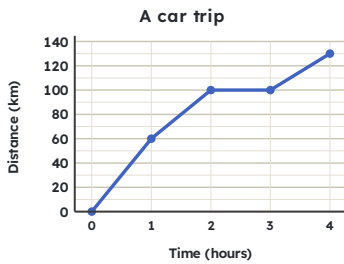
9 A team won 8 games and lost 9. Write the ratio of wins to games played.

from 12.01

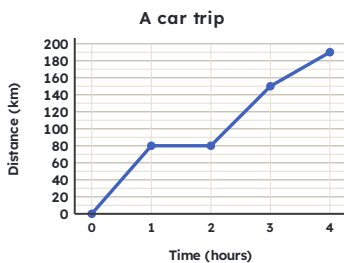
Medium

Questions 10–14

- 10 In which hour was the car fastest? How can you tell?



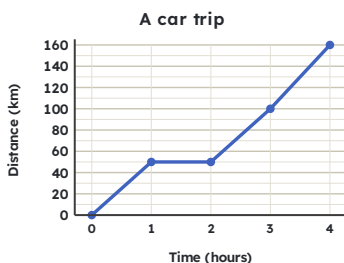
- 11 Find the speed between 2 h and 3 h.



- 12 Find the range of 18, 12, 9, 50, 33, 1, 36.

from 10.06

- 13 Find the average speed for the whole trip.



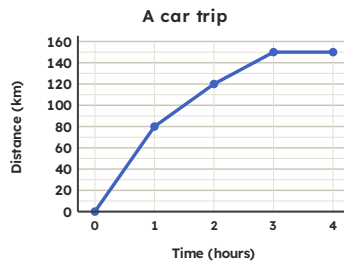
- 14 Find the missing number: $8 : 7 = \square : 21$.

from 12.02

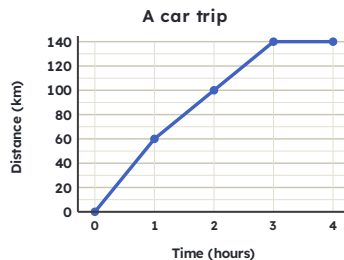
Challenging

Questions 15–18

- 15 Find the average speed for the whole trip.



- 16 About when was the car halfway (70 km)?



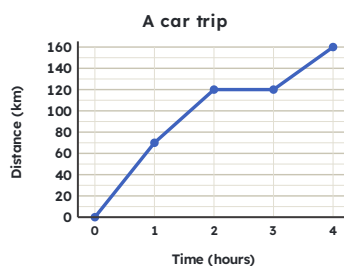
- 17 Find the median.

Stem	Leaf
2	1 1 7
3	
4	1 7
5	1 3 7 9

Key: 2 | 4 = 24

from 10.07

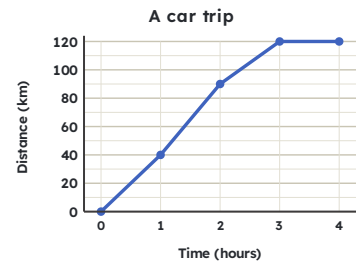
- 18 Find the average speed for the whole trip.



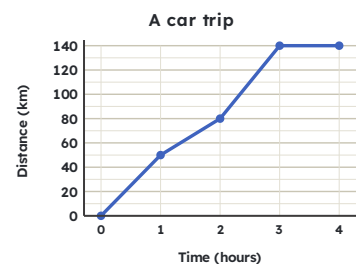
Challenging

Questions 19–23

- 19 About when was the car halfway (60 km)?



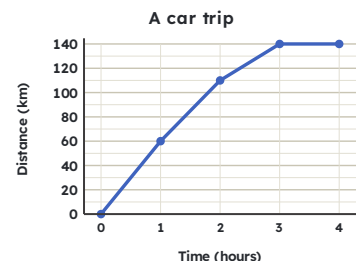
- 20 Find the average speed for the whole trip.



- 21 A die is rolled. Find $P(\text{not } 5)$.

from 11.05

- 22 In which hour was the car fastest? How can you tell?



- 23 In a ratio of 6 : 8, what fraction is the first part?

from 12.01

NAME _____ CLASS _____ DATE _____

WE DO**KEY IDEA**

1 hour = 60 minutes and 1 minute = 60 seconds, so time is not a decimal system: 1.5 h is 1 h 30 min. To add or subtract times, work in hours and minutes, carrying 60 minutes as 1 hour.

Watch your teacher.
Copy every step.

E1 Change 2.25 hours to hours and minutes.

E2 Add 2 h 45 min and 1 h 30 min.

E3 Change 200 minutes to hours and minutes.

Easy

Questions 1–9

1 How many minutes are in 5 hours?

2 Change 153 minutes to hours and minutes.

3 Change 1.1 hours to hours and minutes.

4 How many minutes are in 6 hours?

5 Change 242 minutes to hours and minutes.

6 Change 2.25 hours to hours and minutes.

7 How many minutes are in 1 hour?

8 Change 159 minutes to hours and minutes.

9 Change 50 000 kg to t.

from 8.01

Easy

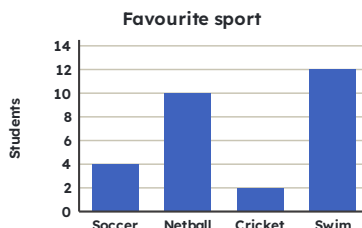
Questions 10–16

10 Change 193 minutes to hours and minutes.

11 Change 3.75 hours to hours and minutes.

12 How many minutes are in 4 hours?

13 Which sport was the most popular?



from 10.01

14 Add 2 h 46 min and 1 h 2 min.

___ h ___ min

= ___ h ___ min

15 Change 104 minutes to hours and minutes.

16 Change 3.1 hours to hours and minutes.

Medium

Questions 17–23

17 Add 2 h 7 min and 3 h 4 min.

18 Subtract: 5 h 23 min – 1 h 54 min.

19 A team won 14 games and lost 4. Write the ratio of wins to games played.

from 12.01

20 Write 1 h 28 min in hours, as a decimal (2 decimal places).

21 Add 3 h 12 min and 2 h 52 min.

22 A tap runs at 25 L/min. How long does it take to fill a 225 L tank?

from 12.06

23 Subtract: 2 h 31 min – 59 min.

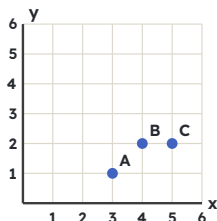
Medium

Questions 24–33

- 24 Subtract: 4 h 31 min – 32 min.

- 25 Write 1 h 2 min in hours, as a decimal (2 decimal places).

- 26 Which point is at (3, 1)?



from 9.02

- 27 Add 35 min and 2 h 5 min.

- 28 Find the missing number: 4 : 7 = □ : 21.

from 12.02

- 29 Change 714 seconds to minutes and seconds.

- 30 Subtract: 5 h 6 min – 2 h 8 min.

- 31 2 kg costs \$8.00 or 3 kg costs \$10.50. Find the price per kg of each.

from 12.05

- 32 Write 3 h 1 min in hours, as a decimal (2 decimal places).

- 33 Change 2 h 33 min to minutes.

from 8.01

Challenging

Questions 34–40

- 34 Change 657 seconds to minutes and seconds.

- 35 5 lessons each last 43 minutes. How long is that in hours and minutes?

- 36 Find the median.

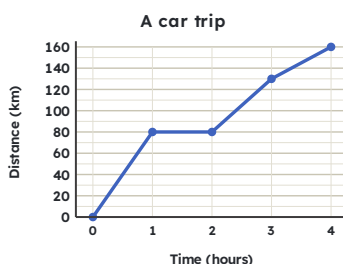
Stem	Leaf
2	2 5 9
3	2 7
4	0 0 5
5	1

Key: 2 | 4 = 24

from 10.07

- 37 Mia says 4.5 hours is 4 h 50 min. Is that right? Explain.

- 38 How far from home is the car after 2 hours?



from 12.07

- 39 Change 734 seconds to minutes and seconds.

- 40 3 lessons each last 44 minutes. How long is that in hours and minutes?

Challenging

Questions 41–50

- 41 5 lessons each last 34 minutes. How long is that in hours and minutes?

- 42 Change 290 seconds to minutes and seconds.

- 43 Find the range of 29, 4, 33, 29, 43.

from 10.06

- 44 Write 2 h 32 min in hours, as a decimal (2 decimal places).

- 45 5 kg of apples cost \$30. Find the cost per kilogram.

from 12.04

- 46 5 lessons each last 47 minutes. How long is that in hours and minutes?

- 47 Change 728 seconds to minutes and seconds.

- 48 4 pens cost \$0.96. Find the cost of 1 pen.

from 12.05

- 49 Write 2 h 49 min in hours, as a decimal (2 decimal places).

- 50 Change 16 cm to mm.

from 8.01

NAME _____ CLASS _____ DATE _____

WE DO
KEY IDEA

24-hour time counts the hours from midnight, so it needs no am or pm: 3:45 pm is 1545. For a pm time from 1 pm, add 12 to the hour. Midnight is 0000.

Watch your teacher.
Copy every step.

E1 Write 3:45 pm in 24-hour time.

E2 Write 0820 in 12-hour time.

E3 Write 2310 in 12-hour time.

Easy

Questions 1–9

1 Write 6:15 am in 24-hour time.

2 Write 2040 in 12-hour time.

3 Write 3:30 pm in 24-hour time.

4 Write 1835 in 12-hour time.

5 Write 2:50 am in 24-hour time.

6 Write 0835 in 12-hour time.

7 Write 2:30 am in 24-hour time.

8 Write 0235 in 12-hour time.

9 Change 193 minutes to hours and minutes.

from 12.08

Easy

Questions 10–18

10 Write 0500 in 12-hour time.

11 Write 4:05 pm in 24-hour time.

12 Write 1402 in 12-hour time.

____ - ____ = ____

____ : ____ pm

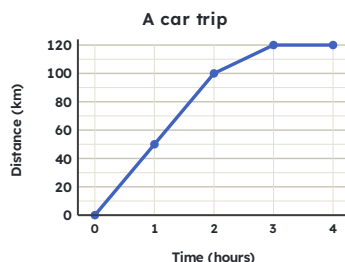
13 A die is rolled. Find P(not 4).

from 11.05

14 Write 1810 in 12-hour time.

15 Write 3:45 pm in 24-hour time.

16 Write 1748 in 12-hour time.

17 During which hour did the car stop?


from 12.07

18 Write 1635 in 12-hour time.

Medium

Questions 19–25

19 Write 1513 in 12-hour time.

20 A train leaves at 1956 and the trip takes 1 h 4 min. When does it arrive? Use 24-hour time.

21 A team won 11 games and lost 15. Write the ratio of wins to games played.

from 12.01

22 Write 1912 in 12-hour time.

23 A train leaves at 1929 and the trip takes 1 h 17 min. When does it arrive? Use 24-hour time.

24 A tap runs at 21 L/min. How long does it take to fill an 84 L tank?

from 12.06

25 Write 2315 in 12-hour time.

Medium

Questions 26–34

- 26 A train leaves at 2206 and the trip takes 1 h 8 min. When does it arrive? Use 24-hour time.

- 27 Write 2013 in 12-hour time.

- 28 A graph has no labels on its axes. Why is that a problem?

from 10.02

- 29 How long is it from 0707 to 1422?

- 30 Find the missing number: $1 : 9 = \square : 72$.

from 12.02

- 31 A train leaves at 1541 and the trip takes 35 min. When does it arrive? Use 24-hour time.

- 32 Write 1616 in 12-hour time.

- 33 Change 4.5 hours to hours and minutes.

from 12.08

- 34 How long is it from 0904 to 1402?

Challenging

Questions 35–42

- 35 How long is it from 1013 to 1917?

- 36 A flight leaves at 2339 and takes 3 h 31 min. When does it land, in 12-hour time?

- 37 How many values are 55 or more?

Stem	Leaf
3	6 6 9
4	3 9
5	0 9
6	8
7	6 7

Key: 3 | 5 = 35

from 10.04

- 38 How long is it from 1018 to 2030?

- 39 Jack earns \$18 an hour. How much is earned in 8 hours?

from 12.06

- 40 A flight leaves at 2102 and takes 2 h 23 min. When does it land, in 12-hour time?

- 41 How long is it from 0737 to 1632?

- 42 Cordial and water are mixed 1 : 4. How much water goes with 200 mL of cordial?

from 12.03

Challenging

Questions 43–50

- 43 A flight leaves at 2303 and takes 1 h 57 min. When does it land, in 12-hour time?

- 44 How long is it from 1135 to 1824?

- 45 A letter is chosen from the word SUMMER. List the sample space.

from 11.01

- 46 A train leaves at 1034 and the trip takes 39 min. When does it arrive? Use 24-hour time.

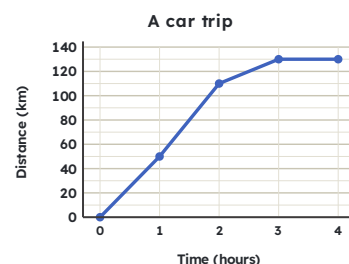
- 47 8 kg of apples cost \$72. Find the cost per kilogram.

from 12.04

- 48 A flight leaves at 2121 and takes 2 h 26 min. When does it land, in 12-hour time?

- 49 How long is it from 0839 to 1505?

- 50 During which hour did the car stop?



from 12.07

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

To find a time difference, count on: up to the next hour, then whole hours, then the minutes left. For ages, subtract the birth year (one less if the birthday has not come yet this year).

E1 How long is it from 9:40 am to 2:15 pm?

E2 Someone was born in 2013. How old are they on their birthday in 2026?

E3 A film starts at 7:25 pm and runs 1 h 50 min. When does it end?

Easy

Questions 1–9

1 How long is it from 7:30 am to 10:00 am?

2 Someone was born in 2010. How old are they on their birthday in 2026?

3 How long is it from 7:15 am to 11:30 am?

4 Someone was born in 2008. How old are they on their birthday in 2026?

5 How long is it from 8:45 am to 1:15 pm?

6 Someone was born in 2015. How old are they on their birthday in 2026?

7 How long is it from 10:15 am to 11:15 am?

8 Someone was born in 2013. How old are they on their birthday in 2026?

9 Write 0100 in 12-hour time.

from 12.09

Easy

Questions 10–18

10 Someone was born in 2011. How old are they on their birthday in 2026?

11 How long is it from 7:45 am to 9:30 am?

12 How long is it from 8:03 am to 11:22 am?

count on to ____:____ am: ____ min

total: ____ h ____ min

13 Set A: 9, 11, 15, 2, 3. Set B: 14, 14, 1, 8, 1. Which set has the higher median?

from 10.08

14 Someone was born in 2009. How old are they on their birthday in 2026?

15 How long is it from 10:00 am to 11:15 am?

16 How long is it from 11:50 am to 3:57 pm?

17 Change 192 minutes to hours and minutes.

from 12.08

18 Someone was born in 2012. How old are they on their birthday in 2026?

Medium

Questions 19–23

19 How long is it from 9:26 am to 11:42 am?

20 A film starts at 5:55 pm and runs 1 h 41 min. When does it end?

21 A team won 14 games and lost 10. Write the ratio of wins to games played.

from 12.01

22 How long is it from 8:22 am to 12:29 pm?

23 A film starts at 8:05 pm and runs 2 h 27 min. When does it end?

Medium

Questions 24–33

- 24 A film starts at 8:55 pm and runs 2 h 3 min. When does it end?
- 25 How long is it from 9:37 am to 3:18 pm?
- 26 Describe the chance of rolling less than 6 on a die.
- 27 Someone was born in July 1977. How old are they in June 2026?
- 28 Find the missing number: $4 : 5 = \square : 30$.
- 29 A film starts at 6:55 pm and runs 1 h 41 min. When does it end?
- 30 How long is it from 11:37 am to 3:52 pm?
- 31 Write 7:30 pm in 24-hour time.
- 32 Someone was born in March 1983. How old are they in June 2026?
- 33 Write 2 h 18 min in hours, as a decimal (2 decimal places).

from 11.03

from 12.02

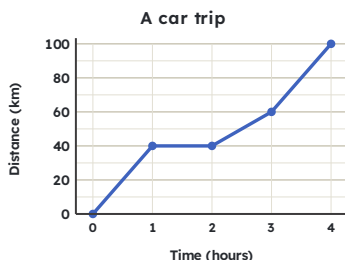
from 12.09

from 12.08

Challenging

Questions 34–41

- 34 Someone was born in March 1984. How old are they in June 2026?
- 35 Lena goes to sleep at 11:30 pm and wakes at 5:00 am. How long is that?
- 36 Start at C5. Move 2 left and 2 down. What is the grid reference now?
- 37 Someone was born in November 1985. How old are they in June 2026?
- 38 How far from home is the car after 3 hours?
- 39 Ruby goes to sleep at 9:15 pm and wakes at 6:00 am. How long is that?
- 40 Someone was born in November 1957. How old are they in June 2026?
- 41 Cordial and water are mixed 1 : 9. How much water goes with 300 mL of cordial?



from 12.07

from 12.03

Challenging

Questions 42–50

- 42 Tom goes to sleep at 10:45 pm and wakes at 5:00 am. How long is that?
- 43 Someone was born in November 1966. How old are they in June 2026?
- 44 A bag has 3 blue and 1 yellow counters. One is picked. Find $P(\text{blue})$.
- 45 A film starts at 8:05 pm and runs 1 h 56 min. When does it end?
- 46 4 kg of apples cost \$36. Find the cost per kilogram.
- 47 Kai goes to sleep at 11:30 pm and wakes at 7:30 am. How long is that?
- 48 Someone was born in July 1986. How old are they in June 2026?
- 49 How many minutes are in 4 hours?
- 50 A film starts at 7:05 pm and runs 2 h 7 min. When does it end?

from 11.02

from 12.04

from 12.08

NAME _____ CLASS _____ DATE _____

WE DO
KEY IDEA

Read a timetable down a column for one trip and across a row for one stop. Find the first time you can catch, then count on to find how long a trip takes.

Watch your teacher.
Copy every step.

E1 When does the first bus reach the Beach?

Station	0700	0745	0830
Library	0712	0757	0842
Hospital	0720	0805	0850
Beach	0736	0821	0906
Mall	0745	0830	0915

E2 How long does the bus take from the Station to the Mall?

E3 You get to the Library at 0730. When is the next bus to the Mall, and when does it arrive?

Easy

Questions 1–4

1 When does the third bus reach the Hospital?

Station	0615	0700	0745
Library	0627	0712	0757
Hospital	0637	0722	0807
Beach	0646	0731	0816
Mall	0652	0737	0822

2 When does the third bus reach the Hospital?

Station	0700	0745	0830
Library	0708	0753	0838
Hospital	0714	0759	0844
Beach	0725	0810	0855
Mall	0737	0822	0907

3 When does the first bus reach the Library?

Station	0615	0700	0745
Library	0623	0708	0753
Hospital	0633	0718	0803
Beach	0640	0725	0810
Mall	0646	0731	0816

4 When does the second bus reach the Hospital?

Station	0800	0845	0930
Library	0812	0857	0942
Hospital	0818	0903	0948
Beach	0829	0914	0959
Mall	0839	0924	1009

Easy

Questions 5–8

5 When does the second bus reach the Library?

Station	0615	0700	0745
Library	0628	0713	0758
Hospital	0637	0722	0807
Beach	0646	0731	0816
Mall	0657	0742	0827

6 How long does the bus take from the Library to the Mall?

Station	0600	0645	0730
Library	0608	0653	0738
Hospital	0622	0707	0752
Beach	0629	0714	0759
Mall	0642	0727	0812

____ minutes

7 When does the first bus reach the Mall?

Station	0615	0700	0745
Library	0624	0709	0754
Hospital	0637	0722	0807
Beach	0646	0731	0816
Mall	0657	0742	0827

8 Change 4 148 mL to litres.

from 8.13

Medium

Questions 9–12

9 How long does the bus take from the Station to the Mall?

Station	0615	0700	0745
Library	0625	0710	0755
Hospital	0638	0723	0808
Beach	0646	0731	0816
Mall	0655	0740	0825

10 How long does the bus take from the Hospital to the Mall?

Station	0615	0700	0745
Library	0626	0711	0756
Hospital	0639	0724	0809
Beach	0650	0735	0820
Mall	0700	0745	0830

11 A team won 15 games and lost 3. Write the ratio of wins to games played.

from 12.01

12 How long does the bus take from the Library to the Mall?

Station	0800	0845	0930
Library	0809	0854	0939
Hospital	0819	0904	0949
Beach	0832	0917	1002
Mall	0846	0931	1016

Medium

Questions 13–18

- 13 How long does the bus take from the Station to the Mall?

Station	0800	0845	0930
Library	0809	0854	0939
Hospital	0821	0906	0951
Beach	0827	0912	0957
Mall	0834	0919	1004

- 14 You get to the Library at 0830. When does the next bus leave there, and when does it reach the Mall?

Station	0815	0900	0945
Library	0825	0910	0955
Hospital	0835	0920	1005
Beach	0841	0926	1011
Mall	0854	0939	1024

- 15 A letter is chosen from the word PENCIL. List the sample space.

from 11.01

- 16 How long does the bus take from the Station to the Mall?

Station	0700	0745	0830
Library	0710	0755	0840
Hospital	0716	0801	0846
Beach	0730	0815	0900
Mall	0738	0823	0908

- 17 Find the missing number: $1 : 3 = \square : 15$.

from 12.02

- 18 You get to the Library at 0711. When does the next bus leave there, and when does it reach the Mall?

Station	0700	0745	0830
Library	0706	0751	0836
Hospital	0716	0801	0846
Beach	0728	0813	0858
Mall	0738	0823	0908

Challenging

Questions 19–23

- 19 You get to the Library at 0831. When does the next bus leave there, and when does it reach the Mall?

Station	0815	0900	0945
Library	0826	0911	0956
Hospital	0840	0925	1010
Beach	0854	0939	1024
Mall	0901	0946	1031

- 20 You must be at the Mall by 0852. Which bus should you catch from the Station?

Station	0715	0800	0845
Library	0728	0813	0858
Hospital	0742	0827	0912
Beach	0754	0839	0924
Mall	0801	0846	0931

- 21 Find the median and the range of 13, 58, 42, 46, 37, 34, 33.

from 10.06

- 22 You get to the Library at 0827. When does the next bus leave there, and when does it reach the Mall?

Station	0815	0900	0945
Library	0822	0907	0952
Hospital	0836	0921	1006
Beach	0843	0928	1013
Mall	0849	0934	1019

- 23 Change 2.75 hours to hours and minutes.

from 12.08

Challenging

Questions 24–28

- 24 You must be at the Mall by 0722. Which bus should you catch from the Station?

Station	0600	0645	0730
Library	0608	0653	0738
Hospital	0622	0707	0752
Beach	0628	0713	0758
Mall	0636	0721	0806

- 25 You get to the Library at 0619. When does the next bus leave there, and when does it reach the Mall?

Station	0600	0645	0730
Library	0614	0659	0744
Hospital	0624	0709	0754
Beach	0636	0721	0806
Mall	0648	0733	0818

- 26 How many values are in the plot?

Stem	Leaf
1	9
2	1 4
3	1 2 8
4	0 2 5 7

Key: 2 | 5 = 25

from 10.04

- 27 How long does the bus take from the Station to the Mall?

Station	0815	0900	0945
Library	0822	0907	0952
Hospital	0828	0913	0958
Beach	0838	0923	1008
Mall	0845	0930	1015

- 28 6 kg of apples cost \$42. Find the cost per kilogram.

from 12.04